

VENKATA GANJI

AI PLATFORM SRE | GPU MODEL SERVING & KUBERNETES
DISTRIBUTED SYSTEMS | INGESTION, RETRIEVAL & RELIABILITY



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linkedin.com/in/venkata-sai-ganji-3a30a2a4 | Architecture Portfolio <https://offroadguy.github.io/>

AI infrastructure and distributed systems engineer with 12 years of production experience.

Across Apple and Freddie Mac engagements and startup environments: shipped production AI tooling, led critical migrations, and owned platform reliability through launch, incident recovery, and verification.

ENTERPRISE AI PLATFORM AND MY CONTRIBUTION

PLATFORM SCOPE Enterprise chat Document intelligence Agent applications Studio UIs & APIs		
OWNED ONBOARDING AND PRODUCTION OPERATIONS		
Ingestion & documents Connectors, parsing, chunking, proprietary document conversion	Retrieval & agents Search, embeddings, memory, orchestration, MCP/tools, guardrails	GPU model serving Open-weight models, vLLM, weight staging, LoRA adapters
OWNED DATA OPERATIONS Vector DBs (Solr, Milvus, OpenSearch), Redis, MongoDB, Elasticsearch & Kafka		
OWNED PLATFORM OPERATIONS Kubernetes, GitOps/IaC, ops tooling, metrics/logs, dashboards/alerts, capacity, cost & DR		
OWNED GPU serving infrastructure, deployment tooling and model-weight onboarding		
Platform footprint: AWS GCP Alibaba Cloud Apple private cloud		
250+ nodes / ~200 TB Vector database infrastructure	~7.0s to ~1.9s Vector-query latency	\$266K+ annualized Solr infrastructure savings

PROFESSIONAL EXPERIENCE



Apple

GenAI Platform Engineering & Cloud Architect

Apr 2024 - Aug 2026

Contract via Mphasis Corp | Enterprise AI infrastructure and platform operations

- Shipped deployment workflows for 10+ GenAI ingestion and inference microservices across Dev/UAT/Prod. Owned Kubernetes/EKS operations, upgrades, routing, autoscaling, resource sizing, and production diagnosis.
- Shipped and owned GPU model-serving and onboarding tooling using Helm/Argo CD and custom vLLM images; configured tensor parallelism, GPU memory budgets, context limits, and LoRA. Built S3-to-FSx weight staging with checksum validation and read-only serving mounts.
- Owned vector database capacity and reliability across Solr, Milvus, and OpenSearch; scaled to **250+ nodes / ~200 TB** and cut query latency from **~7.0s to ~1.9s** through JVM/GC, OS, cache, query, and index tuning.
- Owned database and messaging operations across vector stores, Redis, MongoDB, Elasticsearch, and Kafka. Shipped Solr IaC and lifecycle tooling with Terraform, CloudFormation, Ansible, and Python for provisioning, health checks, backups, recovery, and cleanup.
- Owned observability from metrics and logs to dashboards and alerts; helped reduce incidents from **~20/day to ~2/week** through automation, performance tuning, and recurring-failure remediation.
- Migrated 200 Solr nodes to AWS Graviton/ARM64, preserving throughput while saving **19.8% (\$266K+ annualized)**; productionized EC2 Mac document conversion with application developers, saving **~\$16K/month**.

CLOUDWICK TECHNOLOGIES INC | APR 2014 - APR 2024

Amgen
 AWS Cloud Solutions & Migration Engineer

Sep 2023 - Apr 2024

Cloudwick client engagement

- Migrated **100+ AWS accounts** to Control Tower and Organizations using repeatable Python, CloudFormation, CDK, and SSM workflows; implemented secure S3 migrations with IAM/KMS.
- Tuned vector databases, resolved GenAI ingestion failures, and built Kubernetes deployment pipelines and Python monitoring integrations.

Freddie Mac
 AWS Cloud Solutions Architect & Infrastructure Consultant

Jan 2019 - Aug 2023

Cloudwick client | Technical leadership for 3-8 engineers across 5+ projects

ARCHITECTURE	SOFTWARE ENGINEERING	PRODUCTION OPERATIONS
Rearchitecture & cloud migration	Spark, Python & ingestion pipelines	IaC, performance & disaster recovery

- **Architecture and migration.** Led three mission-critical migrations across distributed data and live OLTP/web platforms; rearchitected mortgage-document systems from Hadoop/Solr to Elasticsearch/OpenSearch and S3.
- **Software engineering.** Led Spark migration-job development for Solr-to-Elasticsearch and HBase-to-S3; designed Databricks, AWS Glue, Talend, and S3 ingestion frameworks and built Python/Airflow automation.
- **Infrastructure and operations.** Built reusable Terraform, CloudFormation, and Ansible provisioning; automated snapshots and instance bootstrapping, supported performance testing, and served as escalation SME for platform and release failures.
- **Disaster recovery.** Delivered a production hybrid-cloud DR environment within six months with ~15-minute RPO and ~2-hour RTO using cross-region replication, S3, Airflow, Python, and Elasticsearch.

Cyber Data Lake / Amorphic
 Lead Solutions Engineer

Nov 2016 - Dec 2018

Cloudwick | Cybersecurity telemetry and distributed analytics

- Built telemetry ingestion correlating PCAP, NetFlow, application logs, and firewall events across Hadoop, Kafka, Spark, Solr, HBase, and Elasticsearch.
- Designed a three-rack IDS/IPS and packet-broker test bed; deployed and tuned 40 Gb/s packet-capture technology and built Prometheus monitoring for collector agents.

EARLIER DISTRIBUTED SYSTEMS DELIVERY**Selected clients | Senior Operations Solutions Engineer | Feb-Oct 2016**

Daimler, SimplyMeasured, Classmates / The Battle Group: Hadoop builds, Ansible automation, security, benchmarking, and production readiness.

Amobee / Turn | Operations Solutions Engineer | Apr 2015-Jan 2016

Operated Hadoop/HBase at up to ~400 nodes per cluster and 30 PB total capacity; delivered HA/DR, fleet refreshes, and Python/JMX monitoring.

Limelight Networks | Operations Engineer | Apr 2014-Apr 2015

Designed HA/DR for Hadoop billing, maintained ~80-node video analytics, and tuned Hive/MapReduce.

TECHNICAL EXPERTISE

AI & orchestration: vLLM, LoRA, Helm, Argo CD, Kubernetes/EKS, Kustomize, Docker, S3, FSx for Lustre

Data & performance: Solr, Milvus, OpenSearch/Elasticsearch, Hadoop/HDFS, HBase, Kafka, Spark; JVM/GC tuning

Automation & reliability: Python, Bash, Terraform, CloudFormation, Ansible, AWS, Prometheus, Grafana, Splunk; HA/DR

EDUCATION

Bachelor of Technology, 2011.

Graduate coursework in Computer Science: Texas A&M University, 15 credits (2014); Montana State University, 15 credits (2013).